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Code

JupyterLab Python 3 (ipykernel)

```
[1]: # to create series
import pandas as pd
a=pd.Series([1,2,3,4,5])
print(a)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
```

```
[11]: import pandas as pd
a=pd.Series([1,2,3,4,5])
print(a)
print(a.values)
print(a.index)
print(a.dtype)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
[1 2 3 4 5]
RangeIndex(start=0, stop=5, step=1)
int64
```

jupyter Day3 Last Checkpoint: 23 hours ago

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JupyterLab Python 3 (ipykernel) ⌵ ☰

RangeIndex(start=0, stop=5, step=1)
int64

```
[18]: #accessing elements
import pandas as pd
a=pd.Series([1,2,3,4,5,6,7])
print(a[1])
#access elements using loc
print(a.iloc[1])
print(a.loc[1])
```

2
2
2

```
[17]: import pandas as pd
a=pd.Series([10,20,30,40,50])
#slicing
print(a[1:3])
print(a.loc[1:3])
print(a.iloc[1:3])
```

```
1    20
2    30
dtype: int64
1    20
2    30
3    40
dtype: int64
```

jupyter Day3 Last Checkpoint: 23 hours ago

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Code

JupyterLab Python 3 (ipykernel)

```
1 20
2 30
3 40
dtype: int64
1 20
2 30
dtype: int64

[57]: #can assign index as we wish
import pandas as pd
a=pd.Series([10,20,30,40,50],index=['A','B','C','D','E'])
print(a)
print(a['E'])
print(a[['A','C','E']])
print(a['A':'D'])
# vectorised operation-no loop required
print(a+5)
print(a.iloc[1]) # can not access through Labeled value
print(a.loc['A']) # can be access through Labeled value

A    10
B    20
C    30
D    40
E    50
dtype: int64
50
A    10
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python 3 (ipykernel)

```
[32]: # statistical functions
import pandas as pd
a=pd.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(a.sum())
print(a.std())
print(a.max())
print(a.min())
print(a.std())
print(a.mean())
print(a.median())
```

```
15
1.5811388300841898
5
1
1.5811388300841898
3.0
3.0
```

```
[26]: import pandas as pd
a=pd.Series([10,20,30,40,50],index=[1,2,3,4,4])
print(a)
print(a[4])
```

```
1 10
2 20
3 30
4 40
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python 3 (ipykernel)

```
[40]: # boolean indexing
import pandas as pd
a=pd.Series([1,2,30,40,50,50,60,3])
print(a[a>3])
# it returns individual value counts
print(a.value_counts())
```

```
2    30
3    40
4    50
5    50
6    60
dtype: int64
50    2
1     1
2     1
30    1
40    1
60    1
3     1
Name: count, dtype: int64
```

```
[54]: #string operation
import pandas as pd
a=pd.Series(['aaa','bb','cc','ddd','ee'])
b=pd.Series(['AAA','BB','CC','DDDD','E'])
print(a.str.upper())
print(a.str.capitalize())
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python 3 (ipykernel)

```
[54]: #string operation
import pandas as pd
a=pd.Series(['aaa','bb','cc','dddd','ee'])
b=pd.Series(['AAA','BB','CC','DDDDD','E'])
print(a.str.upper())
print(a.str.capitalize())
print(b.str.lower())
print(a.str.len())
```

```
0    AAA
1    BB
2    CC
3    DDDD
4    EE
dtype: object
0    Aaa
1    Bb
2    Cc
3    Dddd
4    Ee
dtype: object
0    aaa
1    bb
2    cc
3    ddddd
4    e
dtype: object
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python 3 (ipykernel)

```
dtype: object
0    3
1    2
2    2
3    4
4    2
dtype: int64
```

```
[46]: import pandas as pd
a=pd.Series({"aaa":1,"bb":2,"c":3})
print(a)
print(a["bb"])
```

```
aaa    1
bb     2
c      3
dtype: int64
2
```

```
[53]: #index alignment
import pandas as pd
a=pd.Series([10,20,30],index=['A','B','C'])
b=pd.Series([1,2,3],index=['B','C','D'])
print(a+b)
```

```
A    NaN
B    21.0
C    32.0
D    NaN
```

```
A      NaN
B      21.0
C      32.0
D      NaN
dtype: float64
```

```
[52]: import pandas as pd
a=pd.Series([10,20,30,10,20,40,30,30,50])
print(a.unique())
print(a.count())
print(a.nunique())
```

```
[10 20 30 40 50]
9
5
```

```
[127]: # to create data frame
import pandas as pd
# method1:from dictionary
data={'name':['aaa','bbb','ccc'],
      'rollno':[4,5,6],
      'Age':[18,19,20]}
df=pd.DataFrame(data)
print(df)

# method2:from series to dataframe
a=pd.Series([1,2,3,4,5])
b=pd.Series([10.0,12.4,15.2,5.6,4.8])
c=pd.Series([True,False,True,False])
```


jupyter Day3 Last Checkpoint: 23 hours ago



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Code

JupyterLab Python 3 (ipykernel)

```
# method2:from series to dataframe
a=pd.Series([1,2,3,4,5])
b=pd.Series([10.0,12.4,15.2,5.6,4.8])
c=pd.Series([True,False,True,False])
df=pd.DataFrame({'sno':a,
                 'cgpa':b,
                 'attendance':c})

print(df)

# method3:list of list
data=[['aaa',10,90],
      ['bbb',12,85],
      ['ccc',14,80]]
df=pd.DataFrame(data,columns=['name','age','mark'])
print(df)

# List of dictionary
data=[{'name':'aaa','course':'xx','grade':'A'},
      {'name':'bbb','course':'yy','grade':'O'},
      {'name':'ccc','course':'zz','grade':'A'}]
df=pd.DataFrame(data)
print(df)

data=[{'name':'aaa','course':'xx','grade':'A'},
      {'name':'bbb','course':'yy','grade':'O'},
      {'name':'ccc','course':'zz','grade':'A'}]
df=pd.DataFrame(data,index=['std1','std2','std3'])
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

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Code

JupyterLab Python 3 (ipykernel)

```
data=[{'name':'aaa','course':'xx','grade':'A'},
      {'name':'bbb','course':'yy','grade':'O'},
      {'name':'ccc','course':'zz','grade':'A'}]
df=pd.DataFrame(data,index=['std1','std2','std3'])
print(df)

# adding one extra column
df['marks']=[98,90,80]
print(df)

#adding one extra row
std4=pd.DataFrame([{'name':'ddd','course':'xxx','grade':'O','marks':90}],index=['std4'])
df=pd.concat([df,std4])
print(df)

#adding multiple rows
st=pd.DataFrame([{'name':'eee','course':'yyy','grade':'B','marks':90},
                {'name':'fff','course':'zzz','grade':'O+','marks':90}],index=['std5','std6'])
df=pd.concat([df,st])
print(df)
print("-----")
# column access by bracket notation and dot notation
print(df.course)
print(df['grade'])
print(df[['grade','marks']])

#shape attributes
```

jupyter Day3 Last Checkpoint: 23 hours ago



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Code

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```
#shape attributes
print(df.shape)
#find no of row
print(df.shape[0])
#find no of columns
print(df.shape[1])
#Dtype attribute
#used to check data type of all columns
print(df.dtypes)
#info
print(df.info())
print(df.describe())
#select specific row
print(df.loc['std1'])
print(df.loc[['std1', 'std2']])
print(df.iloc[1])

#update value
df.loc['std3', 'grade'] = 'B++'
print(df)

#delete column
df.drop('grade', axis=1, inplace=True)
print(df)

#delete row
```

```
#delete row
df=df.drop('std4',axis=0)
print(df)

#sorting
df.sort_values('marks',inplace=True)
print(df)
df.sort_values('marks',ascending=False,inplace=True)
print(df)

#statistical function for particular column
print(df['marks'].max())
print(df['marks'].min())
```

```
name  rollno  Age
0  aaa      4   18
1  bbb      5   19
2  ccc      6   20

sno  cgpa  attendance
0   1  10.0         True
1   2  12.4        False
2   3  15.2         True
3   4   5.6        False
4   5   4.8         NaN

name  age  mark
0  aaa  10   90
1  bbb  12   85
2  ccc  14   80
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

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+ Code

JupyterLab Python 3 (ipykernel)

```
print(df['marks'].min())
```

```
name rollno Age
0 aaa      4  18
1 bbb      5  19
2 ccc      6  20
sno  cgpa attendance
0   1  10.0      True
1   2  12.4     False
2   3  15.2      True
3   4   5.6     False
4   5   4.8      NaN
name  age  mark
0  aaa  10   90
1  bbb  12   85
2  ccc  14   80
name course grade
0  aaa    xx    A
1  bbb    yy    0
2  ccc    zz    A
name course grade
std1  aaa    xx    A
std2  bbb    yy    0
std3  ccc    zz    A
name course grade marks
std1  aaa    xx    A    98
std2  bbb    yy    0    90
std3  ccc    zz    A    80
name course grade marks
```

```
std2 bbb yy 0 90
std3 ccc zz A 80
std4 ddd xxx 0 90
name course grade marks
std1 aaa xx A 98
std2 bbb yy 0 90
std3 ccc zz A 80
std4 ddd xxx 0 90
std5 eee yyy B 90
std6 fff zzz 0+ 90
-----
std1 xx
std2 yy
std3 zz
std4 xxx
std5 yyy
std6 zzz
Name: course, dtype: object
std1 A
std2 0
std3 A
std4 0
std5 B
std6 0+
Name: grade, dtype: object
grade marks
std1 A 98
std2 0 90
std3 A 80
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python 3 (ipykernel)

```
(6, 4)
6
4
name      object
course     object
grade      object
marks      int64
dtype: object
<class 'pandas.core.frame.DataFrame'>
Index: 6 entries, std1 to std6
Data columns (total 4 columns):
#   Column  Non-Null Count  Dtype
---  -
0    name    6 non-null      object
1   course  6 non-null      object
2   grade   6 non-null      object
3   marks   6 non-null      int64
dtypes: int64(1), object(3)
memory usage: 240.0+ bytes
None

      marks
count  6.000000
mean   89.666667
std     5.715476
min    80.000000
25%    90.000000
50%    90.000000
75%    90.000000
```

jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

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Code

JupyterLab Python 3 (ipykernel)

```
50% 90.000000
75% 90.000000
max 98.000000
name aaa
course xx
grade A
marks 98
Name: std1, dtype: object
      name course grade marks
std1  aaa     xx     A     98
std2  bbb     yy     0     90
name   bbb
course yy
grade  0
marks  90
Name: std2, dtype: object
      name course grade marks
std1  aaa     xx     A     98
std2  bbb     yy     0     90
std3  ccc     zz    B++     80
std4  ddd     xxx     0     90
std5  eee     yy     B     90
std6  fff     zz     0+     90
      name course marks
std1  aaa     xx     98
std2  bbb     yy     90
std3  ccc     zz     80
std4  ddd     xxx     90
std5  eee     yy     90
```


jupyter Day3 Last Checkpoint: 23 hours ago

File Edit View Run Kernel Settings Help

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Code

JupyterLab Python 3 (ipykernel)

```
std4 ddd xxx 90
std5 eee yyy 90
std6 fff zzz 90
name course marks
std1 aaa xx 98
std2 bbb yy 90
std3 ccc zz 80
std5 eee yyy 90
std6 fff zzz 90
name course marks
std3 ccc zz 80
std2 bbb yy 90
std5 eee yyy 90
std6 fff zzz 90
std1 aaa xx 98
name course marks
std1 aaa xx 98
std2 bbb yy 90
std5 eee yyy 90
std6 fff zzz 90
std3 ccc zz 80
98
80
```

[]: